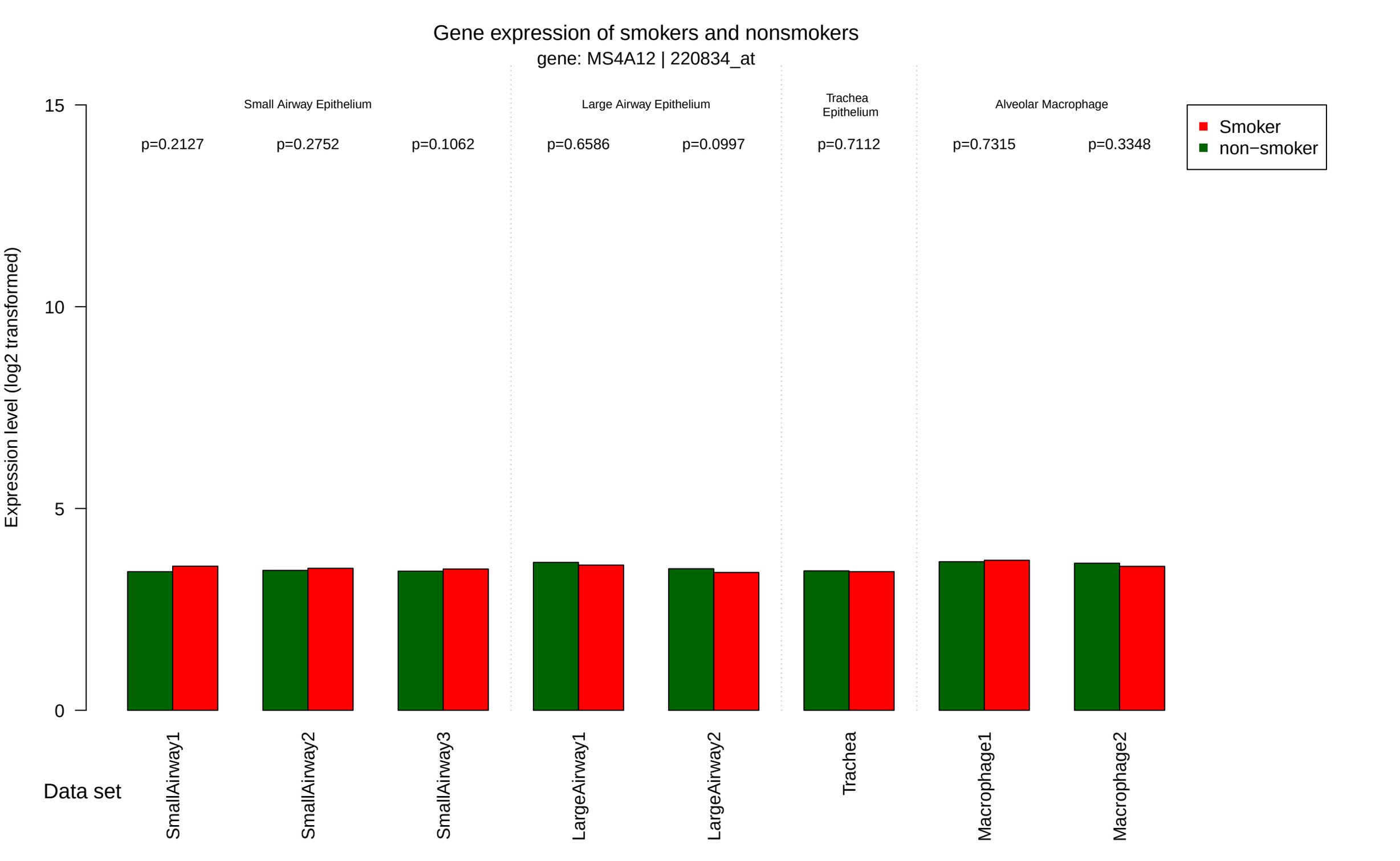
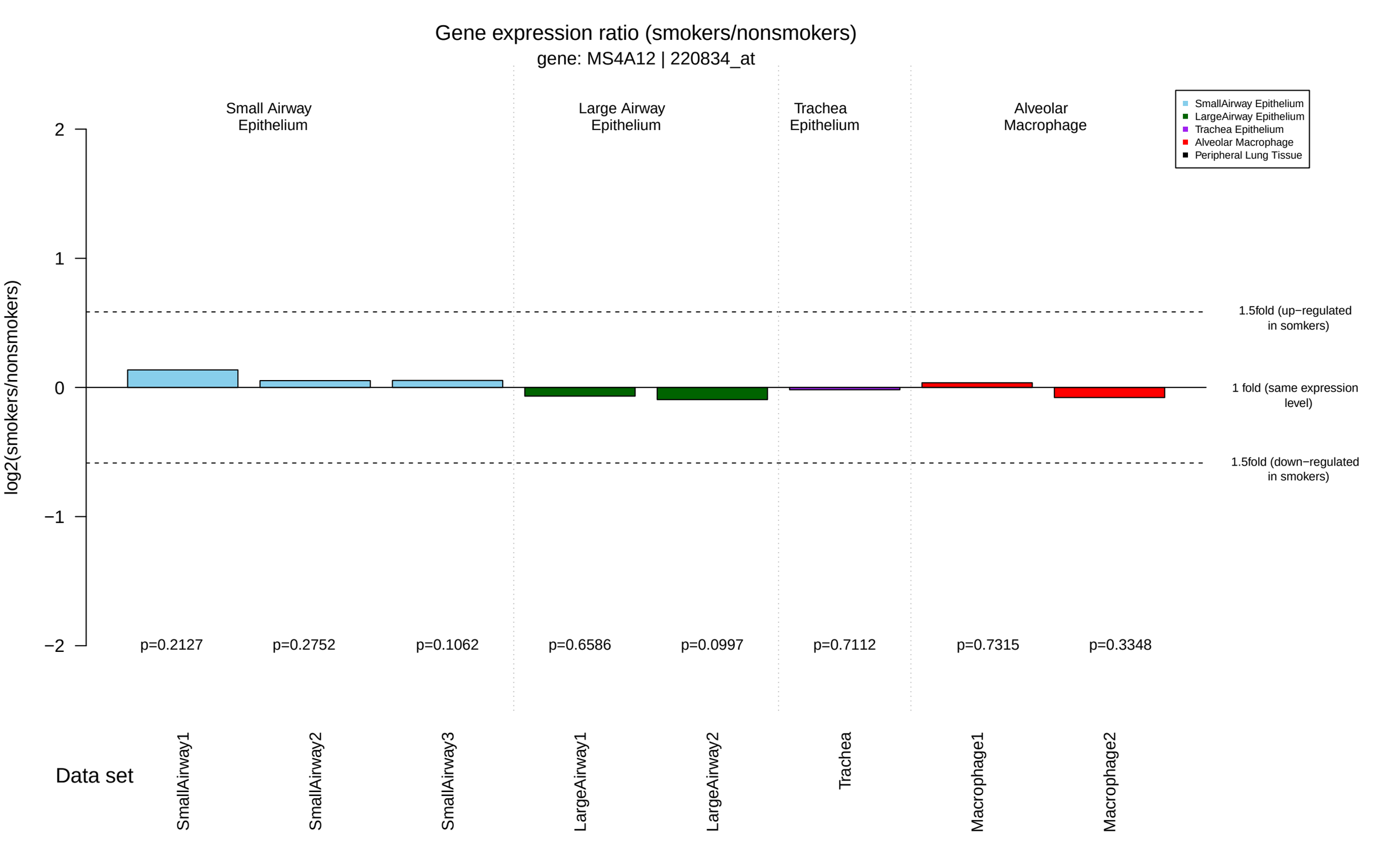
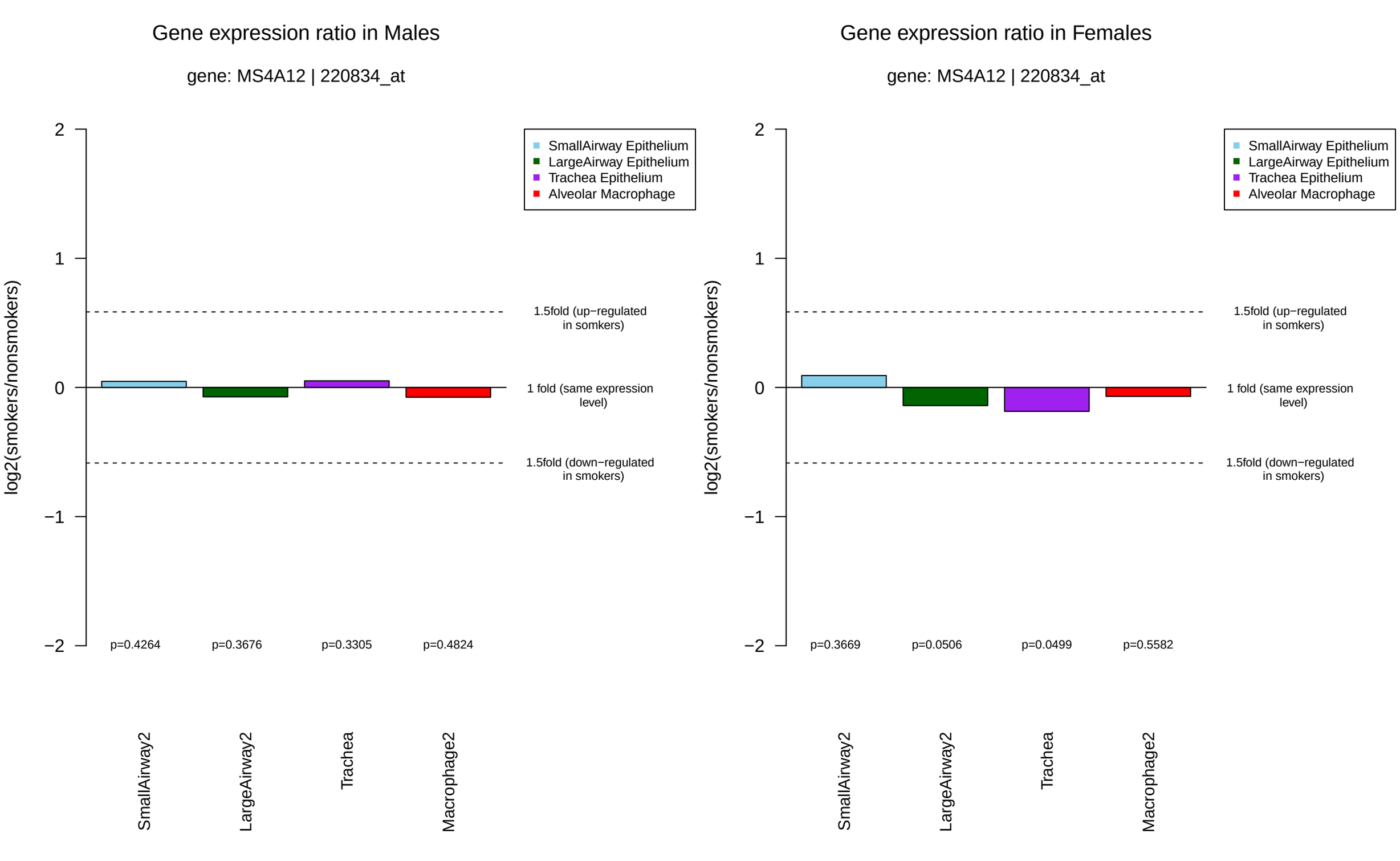
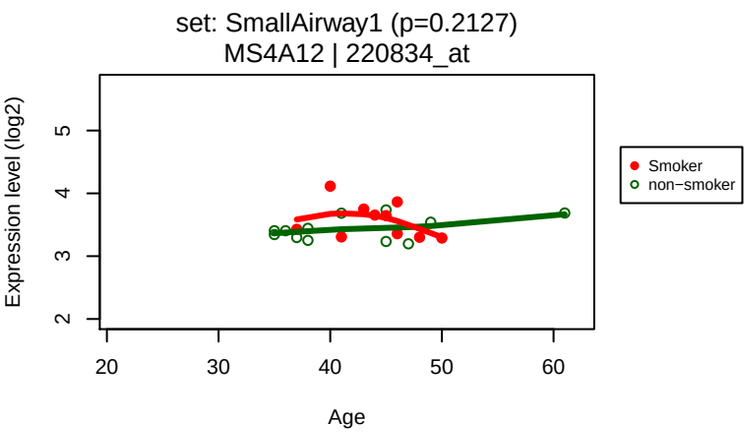
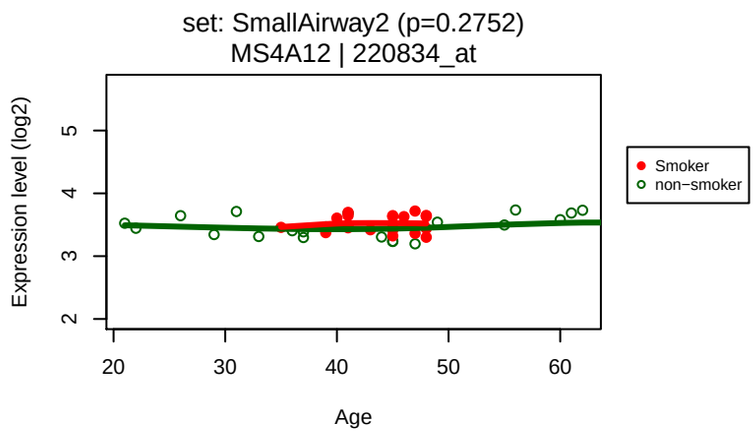
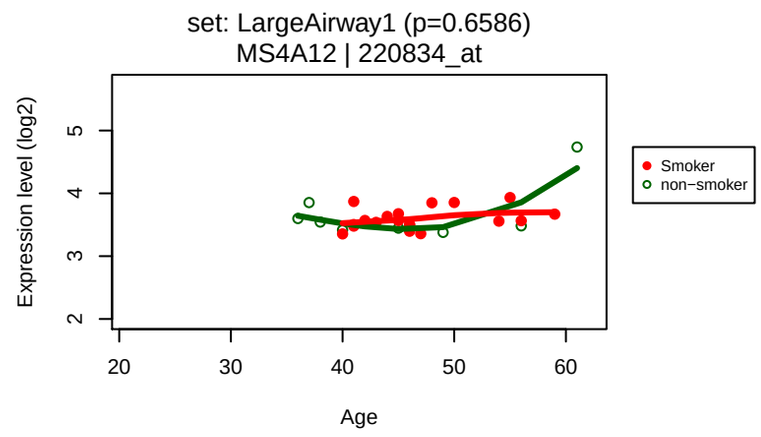

Gene expression of smokers and nonsmokers
gene: MS4A12 | 220834_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2127
p=0.2752
p=0.1062
p=0.6586
p=0.0997
p=0.7112
p=0.7315
p=0.3348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

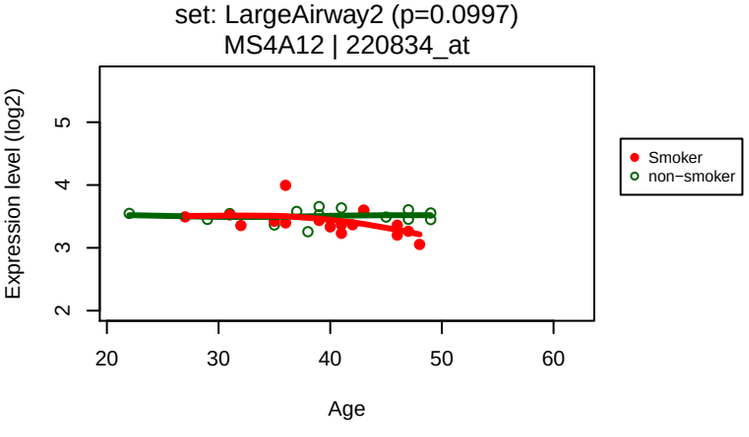

Gene expression ratio (smokers/nonsmokers)
gene: MS4A12 | 220834_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2127
p=0.2752
p=0.1062
p=0.6586
p=0.0997
p=0.7112
p=0.7315
p=0.3348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

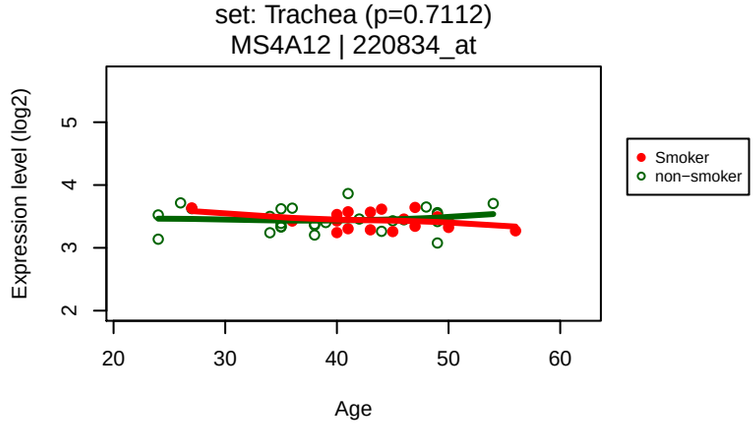

Gene expression ratio in Males
gene: MS4A12 | 220834_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4264
p=0.3676
p=0.3305
p=0.4824
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A12 | 220834_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3669
p=0.0506
p=0.0499
p=0.5582
SmallAirway2
LargeAirway2
Trachea
Macrophage2

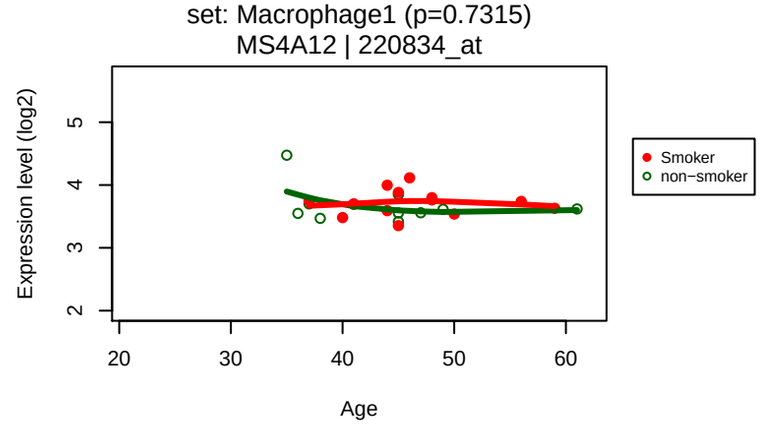

set: SmallAirway1 (p=0.2127)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker

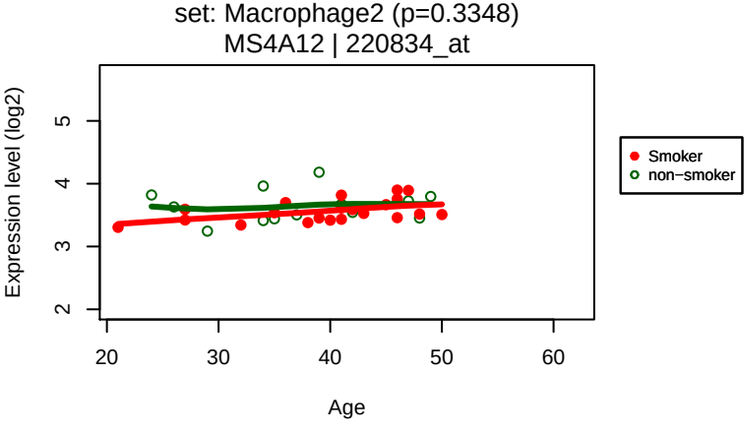

set: SmallAirway2 (p=0.2752)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker

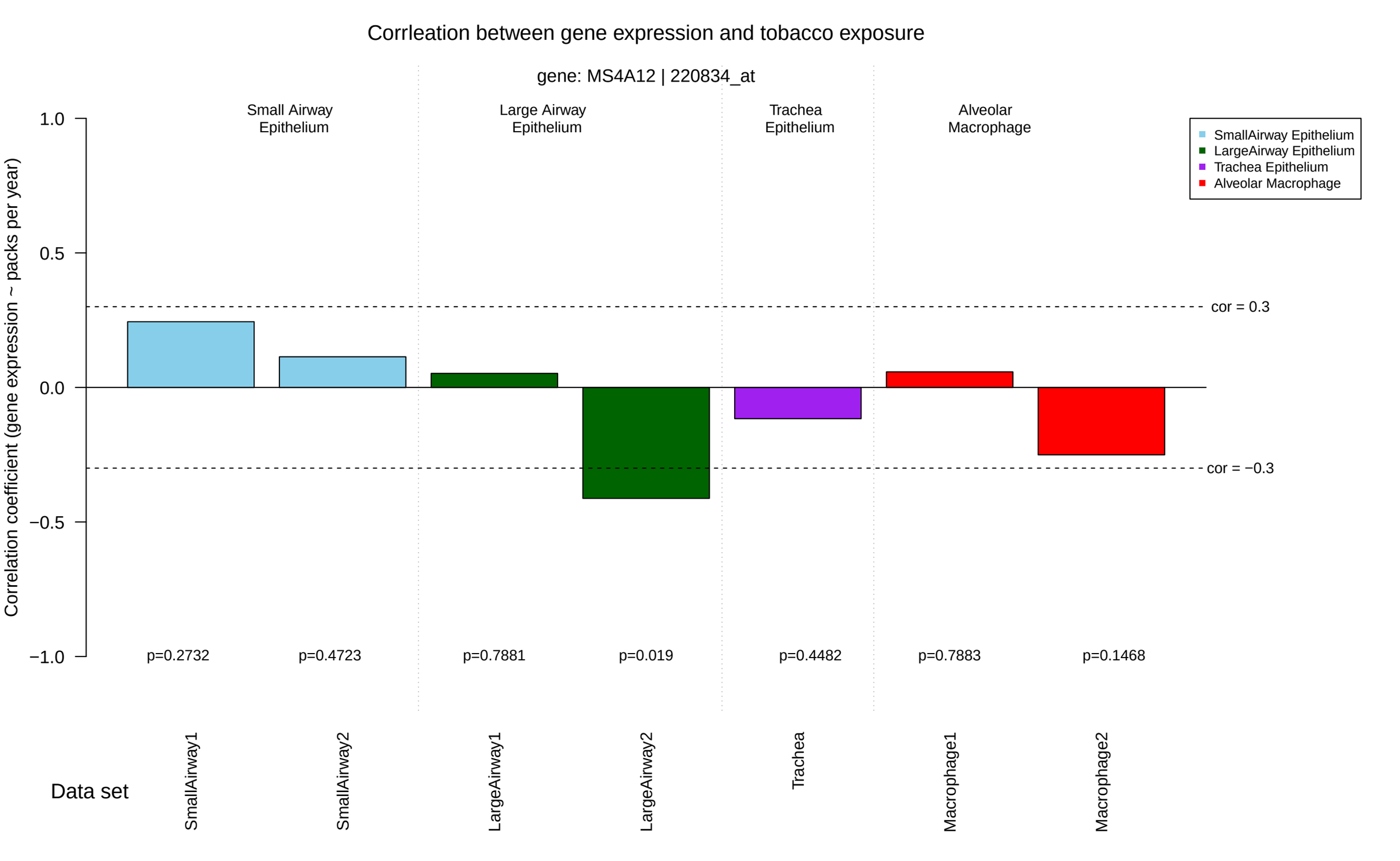

set: LargeAirway1 (p=0.6586)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0997)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7112)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7315)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3348)
MS4A12 | 220834_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A12 | 220834_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2732
p=0.4723
p=0.7881
p=0.019
p=0.4482
p=0.7883
p=0.1468
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2